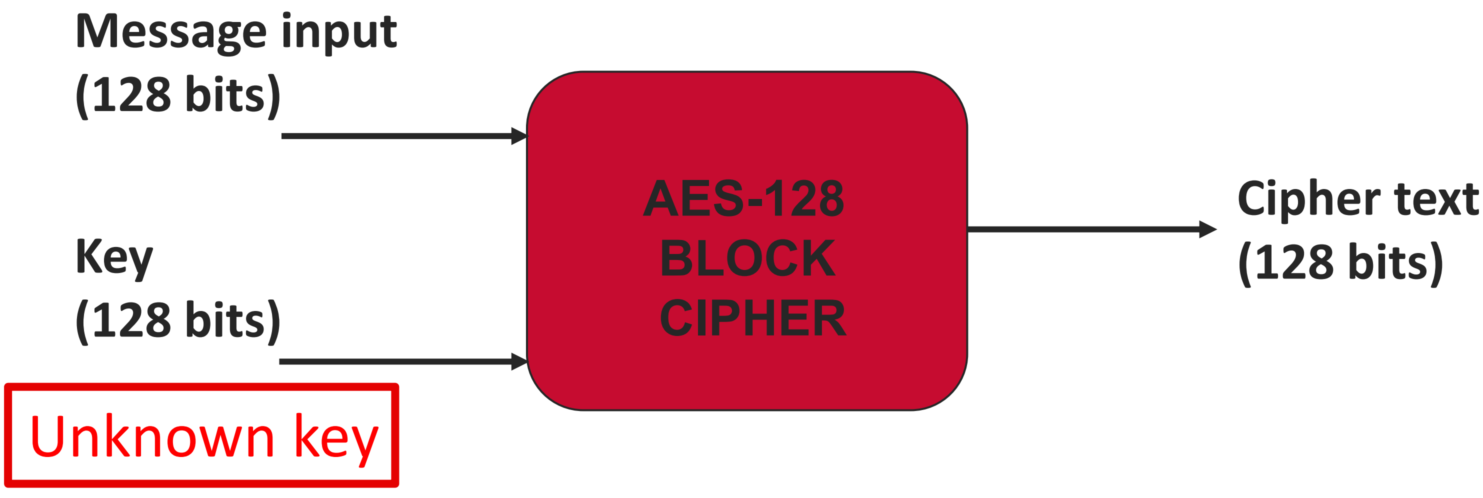
NANYANG TECHNOLOGICAL UNIVERSITY

CE4055

LAB:2

Understanding Correlation Power Analysis

Quick Recap of last session:



- If only given inputs (PT) and outputs (CT), it is computationally not feasible to break the algorithm (irrespective of the number of PT-CT pairs)

- But if we know some **intermediate values**, it is possible to break the encryption algorithm (retrieve the secret key).

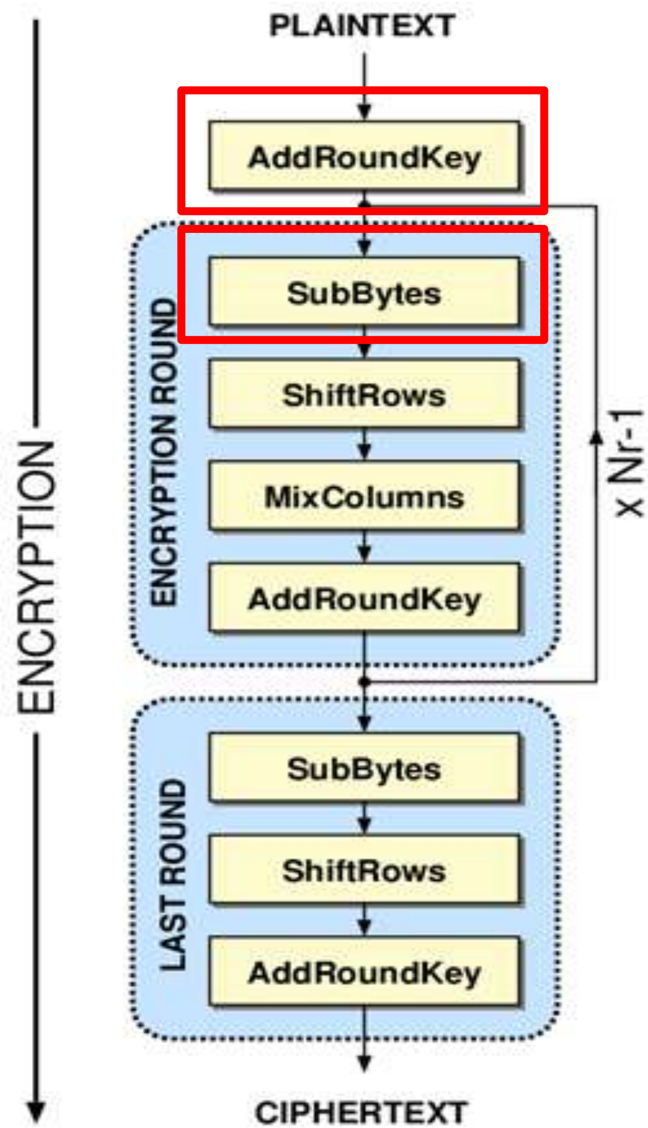
Quick Recap of last session:

- Why measure power-consumption?? **Data is related to power consumption of the device.**

- Measuring the device’s power consumption when it is running the cipher, we get some information about its intermediate values.

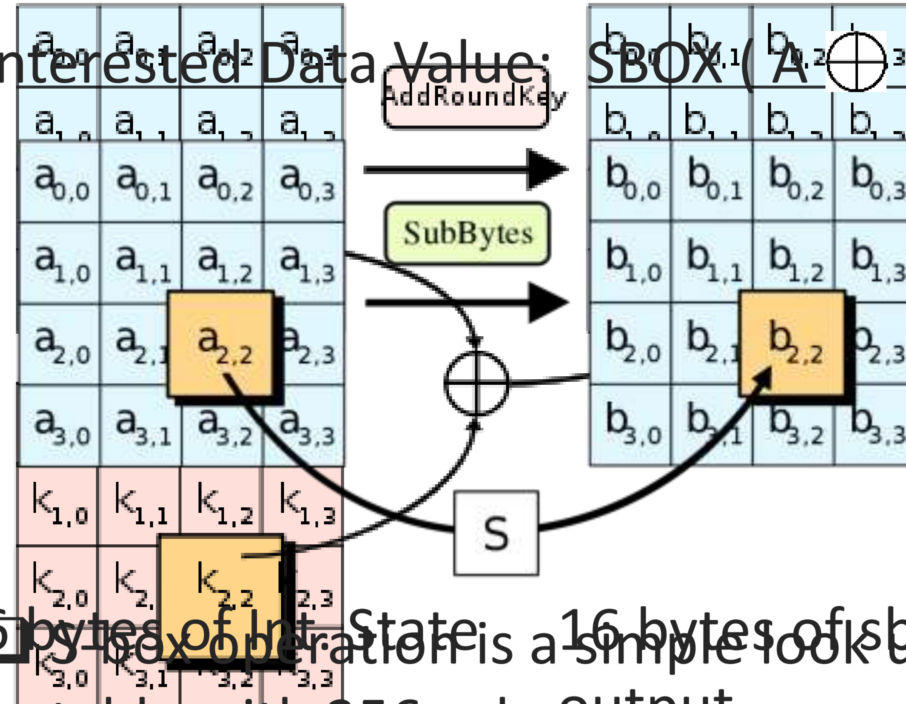
- This might help us find out the secret key used for breaking the cipher.

AES (Advanced Encryption Standard)



16 bytes of PT 16 bytes of Int. state

Interested Data Value: SBOX(A ⊕ K)



16 bytes of Int. State 16 bytes of sbox output

16 bytes of key

- S-box operation is a simple look up table with 256 values.
- The output of SBOX operation is the intermediate value we target for our attack.

AES SBOX:

Input: $a_7a_6a_5a_4a_3a_2a_1a_0$ **Output**: $b_7b_6b_5b_4b_3b_2b_1b_0$

$a_3a_2a_1a_0$

$a_7a_6a_5a_4$

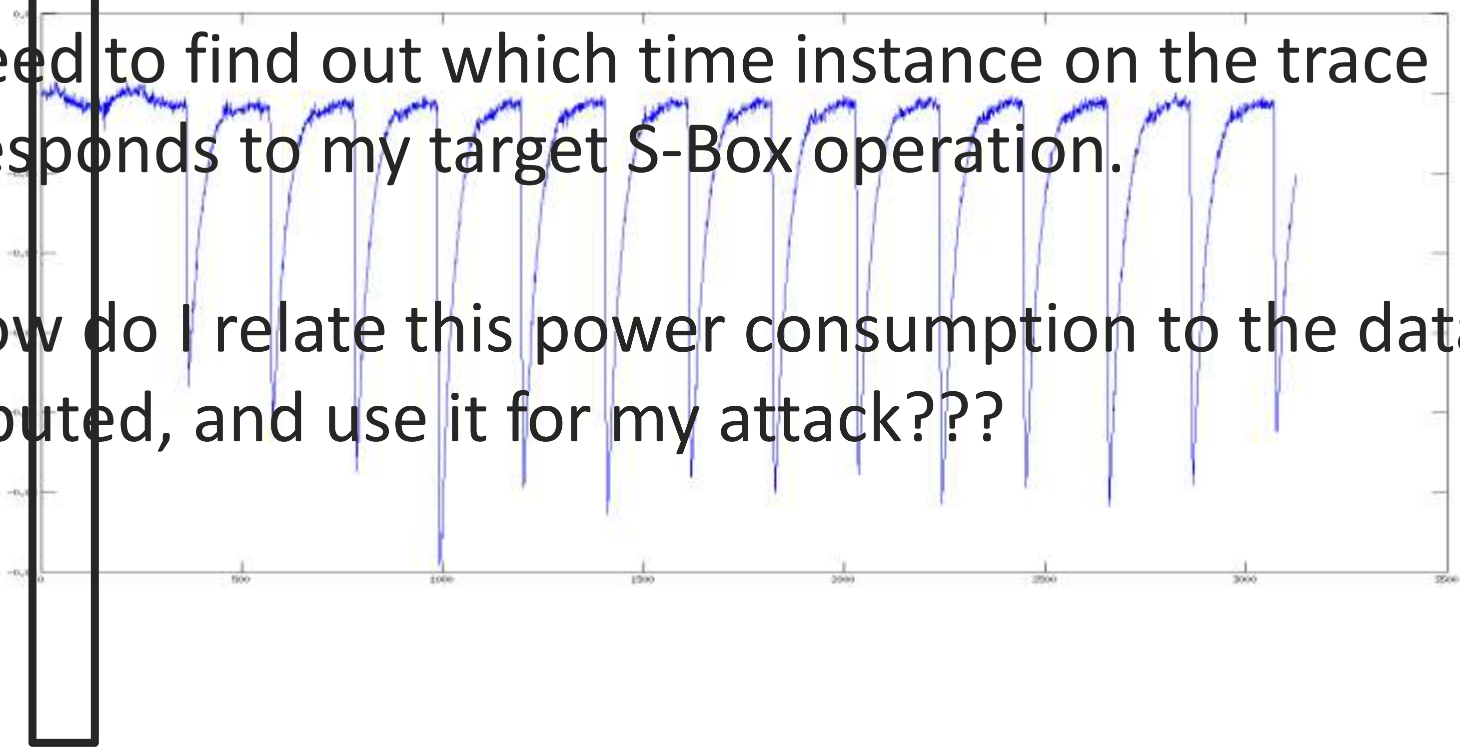
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Power Analysis of AES

$Y = SBOX (A \oplus K)$ Target Value for attack

Natural Questions:

Where is my SBOX operation?

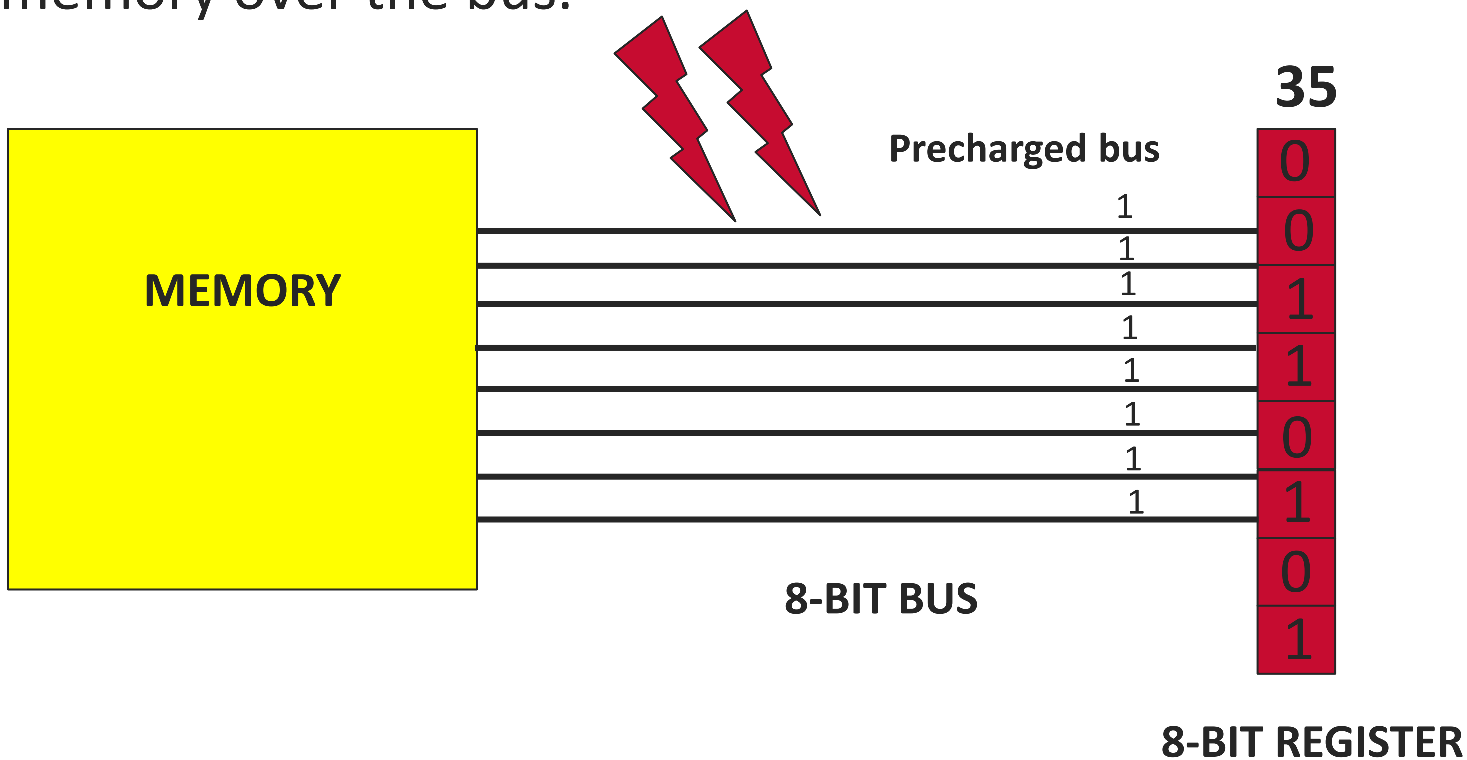


1) Need to find out which time instance on the trace corresponds to my target S-Box operation.

2) How do I relate this power consumption to the data computed, and use it for my attack???

Modelling Power Leakage:

- **SBOX** operation is a memory read operation.
- **Memory Read:** Moving data from register to memory over the bus.



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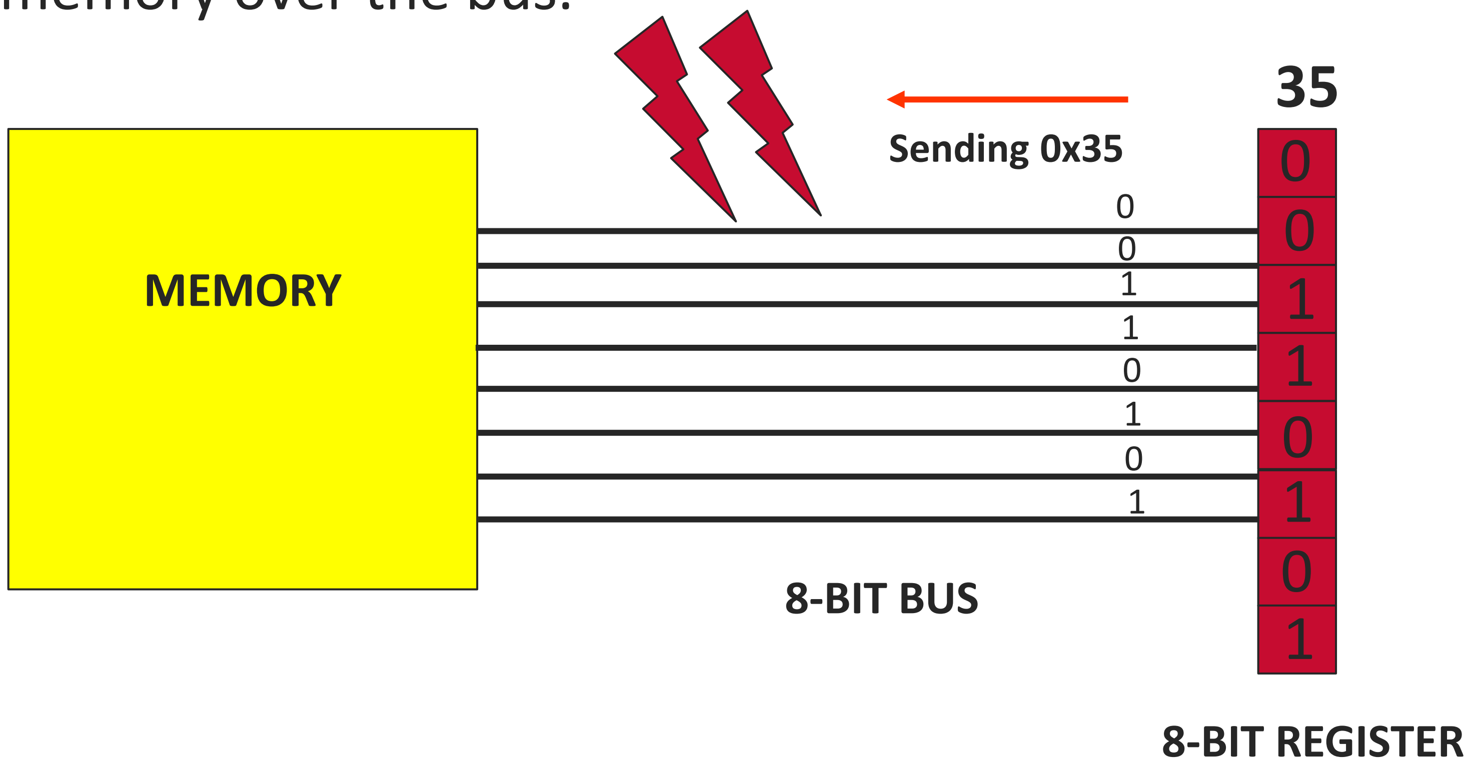

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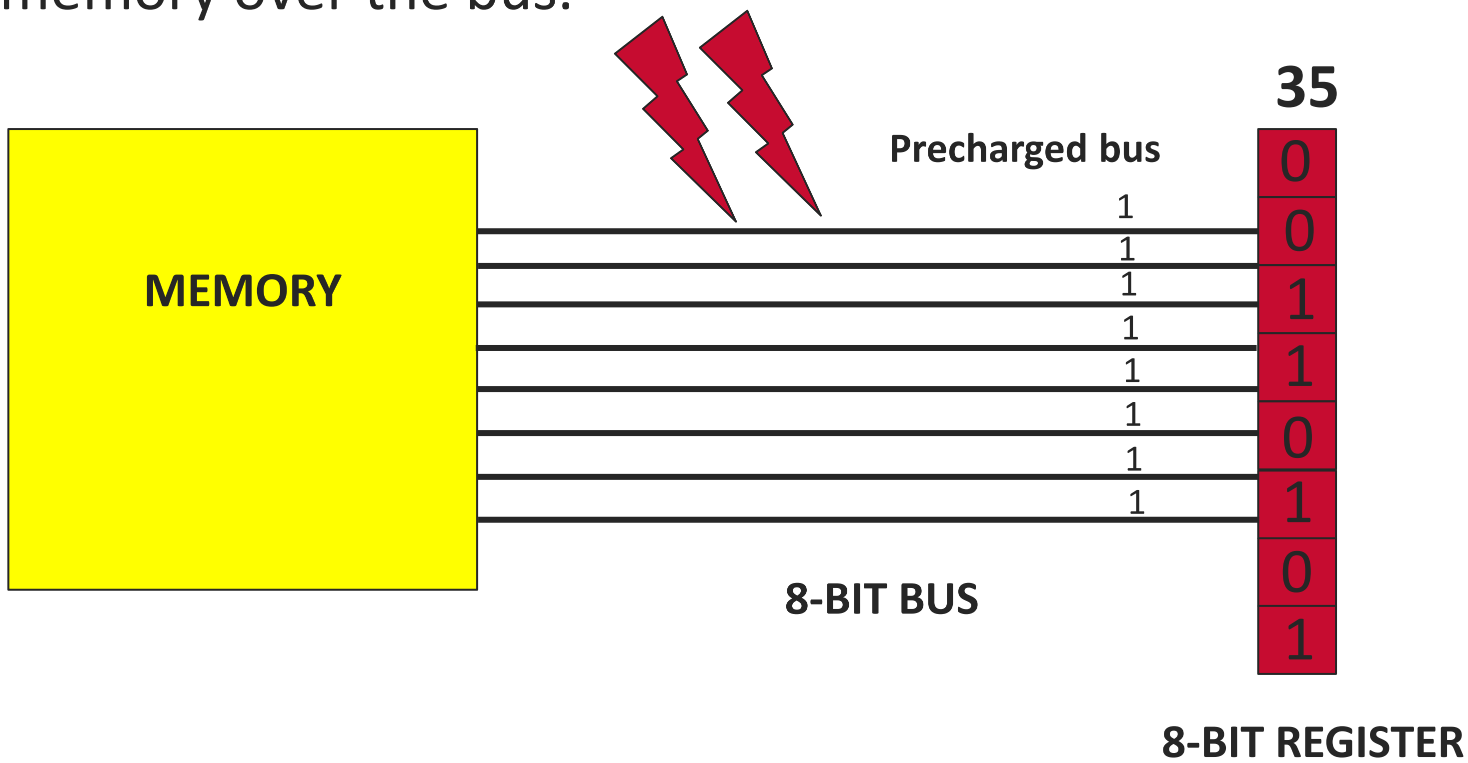

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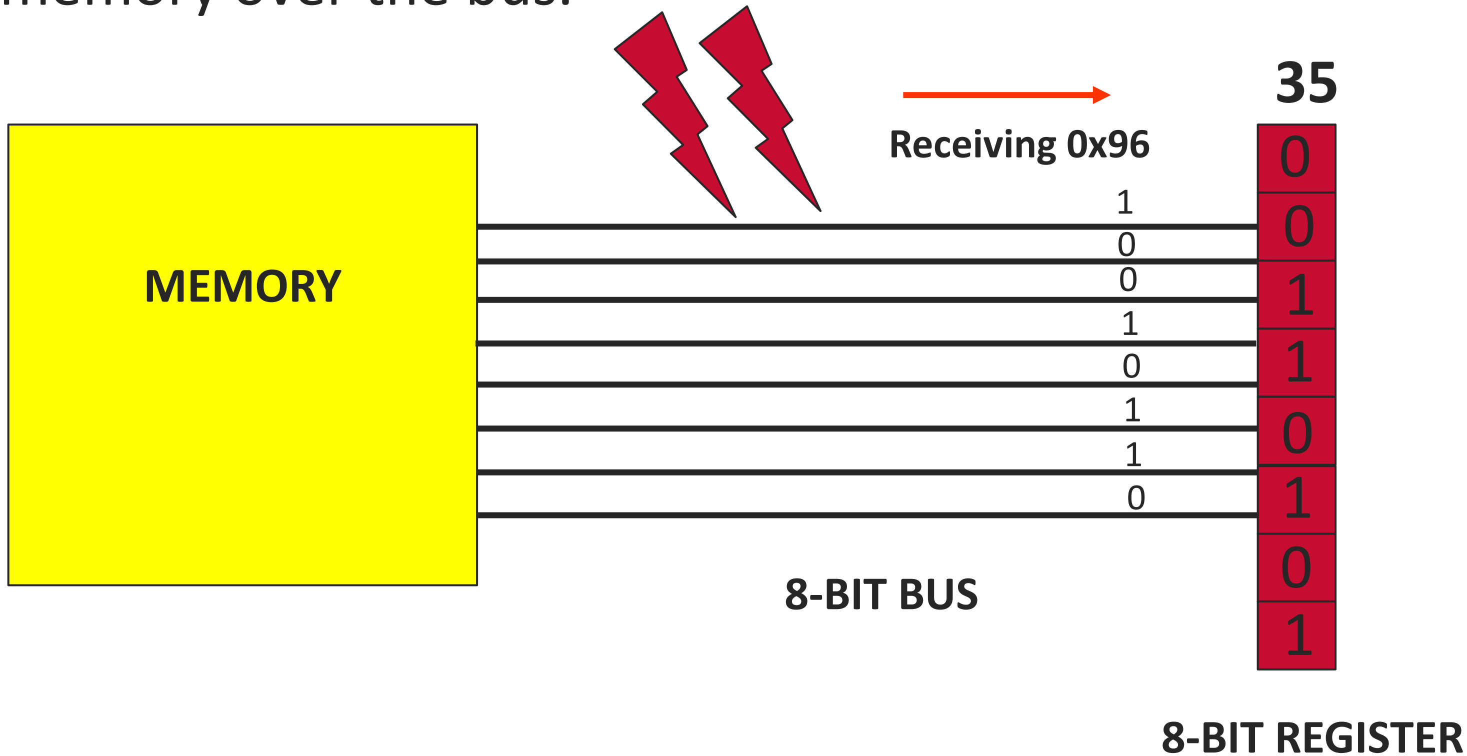

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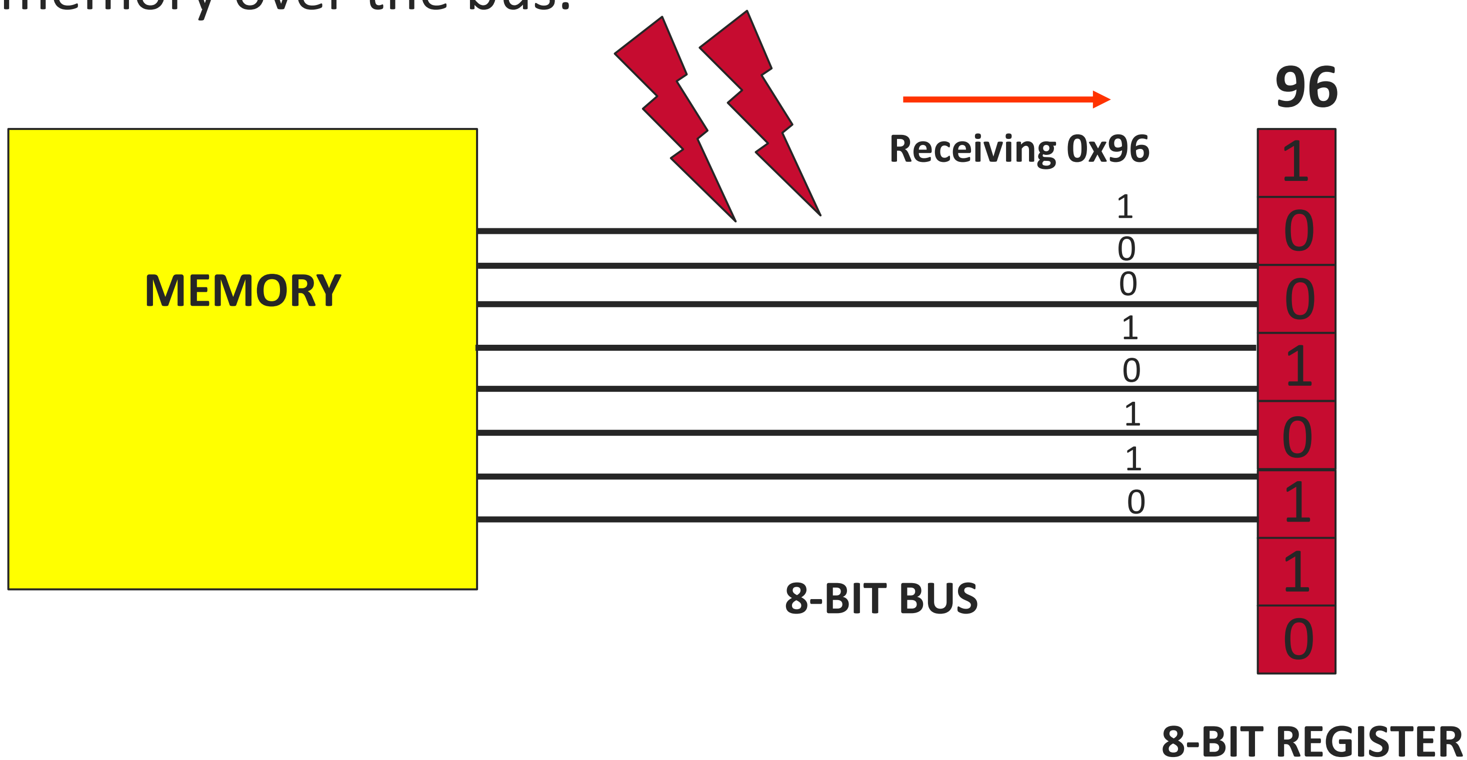

Modelling Power Leakage:

Power Consumed

Initial Bus Value:	1	1	1	1	1	1	1	1
First value switch	0	1	0	1	0	1	0	1

4

4 bits switched !!

LEAKAGE MODEL: HAMMING WEIGHT MODEL

Initial Bus Value:	1	1	1	1	1	1	1	1
Second Value Switch	1	1	0	1	1	1	1	0

2

2 bits switched !!

Power Consumption $\propto$ **Total Number of Bits – Number of 0 bits in value**

HW of value

Number of 1 bits in value

Hamming Weight (HW) of value

Modelling Power Leakage (Exercise)

- To know how to convert data to power…
- Power (A) = Hamming Weight (A)

Data	Power
50 (Decimal)	
99 (Hex)	
889 (Octal)	
ABCDEFG (Hex)	

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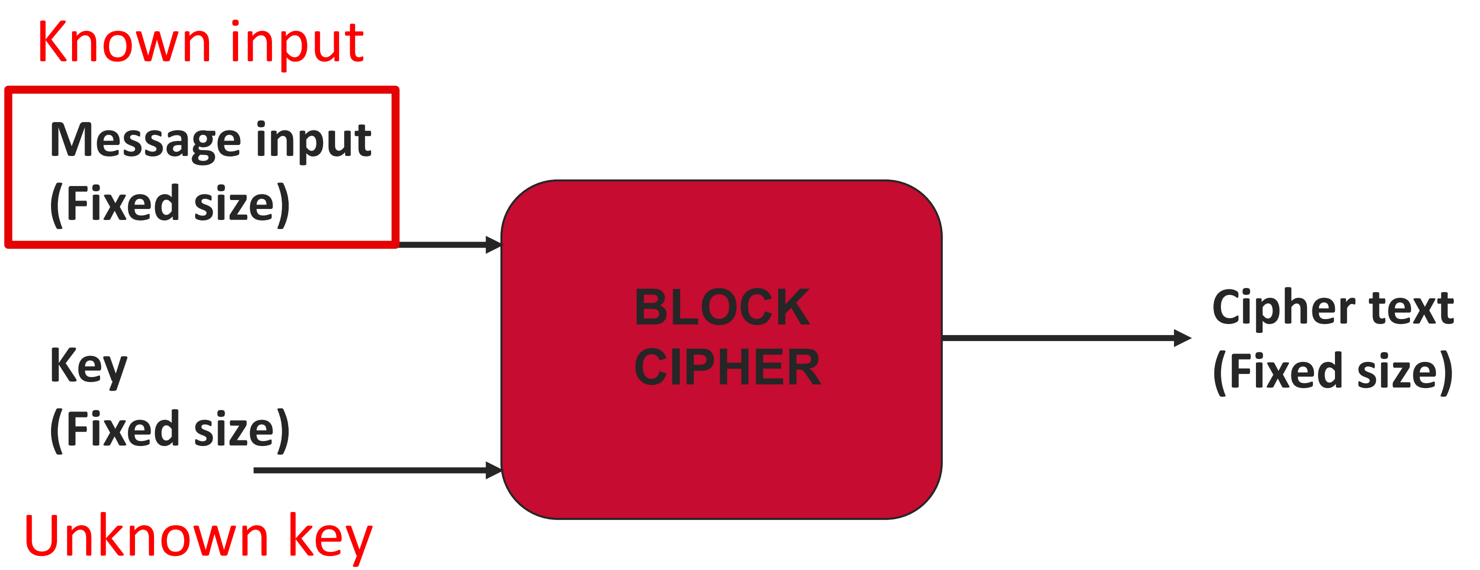
Correlation Power Analysis (CPA)

- Done by observing behaviour of device over many number of operations (Many Traces)

- Is typically performed with known inputs/known outputs.



Correlation Power Analysis (CPA)

- **Divide and Conquer** strategy
  - Retrieve the secret key one byte at a time
  - Repeat the process 16 times for whole key

- **ATTACK PHILOSOPHY (For Each Key Byte):**
  - For every possible value of the key byte, build a "**MODEL**" trace
  - **Match the model** of every possible key byte value with the real traces
  - The model **which matches the best** with the real traces corresponds to the correct key byte.

- What do we model?
  - We model the Output of SBox Operation
  - We use the Hamming Weight Model

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Correlation Power Analysis (CPA)

Lets first concentrate on the first byte of the key.

$Y0 = SBOX(X0 \oplus K0)$ Target Value for attack

1) For K0 = 0x20, calculate set of Y0 for given X0

2) Open https://trinket.io/embed/python3

3) Then, calculate HW(Y0)

4) This is the power model for K0 = 0x20

5) Similarly, we can build model for all possible values of K0 = 0x00 to 0xFF (256 models)

Correlation Power Analysis (CPA)

Lets first concentrate on the first byte of the key.

$Y0 = SBOX(X0 \oplus K0)$ Target Value for attack

1) A single model is a vector

2) What is the length of the vector
   Length: 100

3) For all 256 models, we get a power model matrix (256 x 100)

Correlation Power Analysis (CPA)

1) Now, we match (correlate) model for each byte with the actual power trace.

2) Size of Actual Power trace: 100 x 2500

3) But, where in my trace is the Sbox operation?

4) One Column in this trace will be the target Sbox Operation

5) **Since we do not know which column, we correlate model trace with all columns of actual trace**

Correlation Power Analysis (CPA)

1) **Model Power Trace (**256 x 100)

2) **Actual Power Trace** (100 x 2500)

3) Column-wise correlation of the two matrices.

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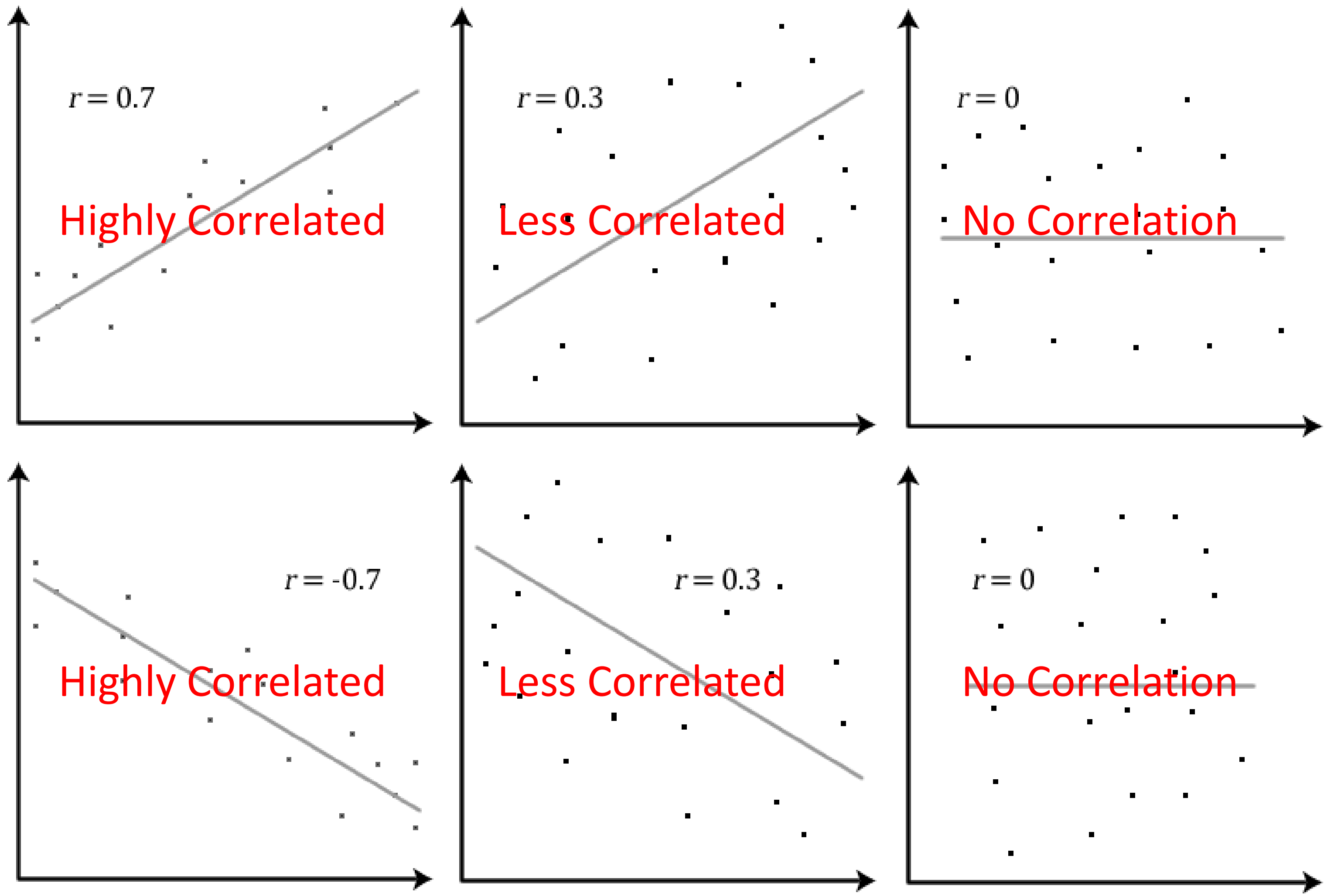
Correlation Power Analysis (CPA)



Correlation Power Analysis (CPA)

1) **Model Power Trace (**256 x 100)

2) **Actual Power Trace** (2500 x 100)

3) Column-wise correlation of the two matrices.

4) You get a correlation matrix (256 x 2500).

5) A single peak at exactly the correct key byte and the correct time instance.

6) **Killing two birds with a single stone…**

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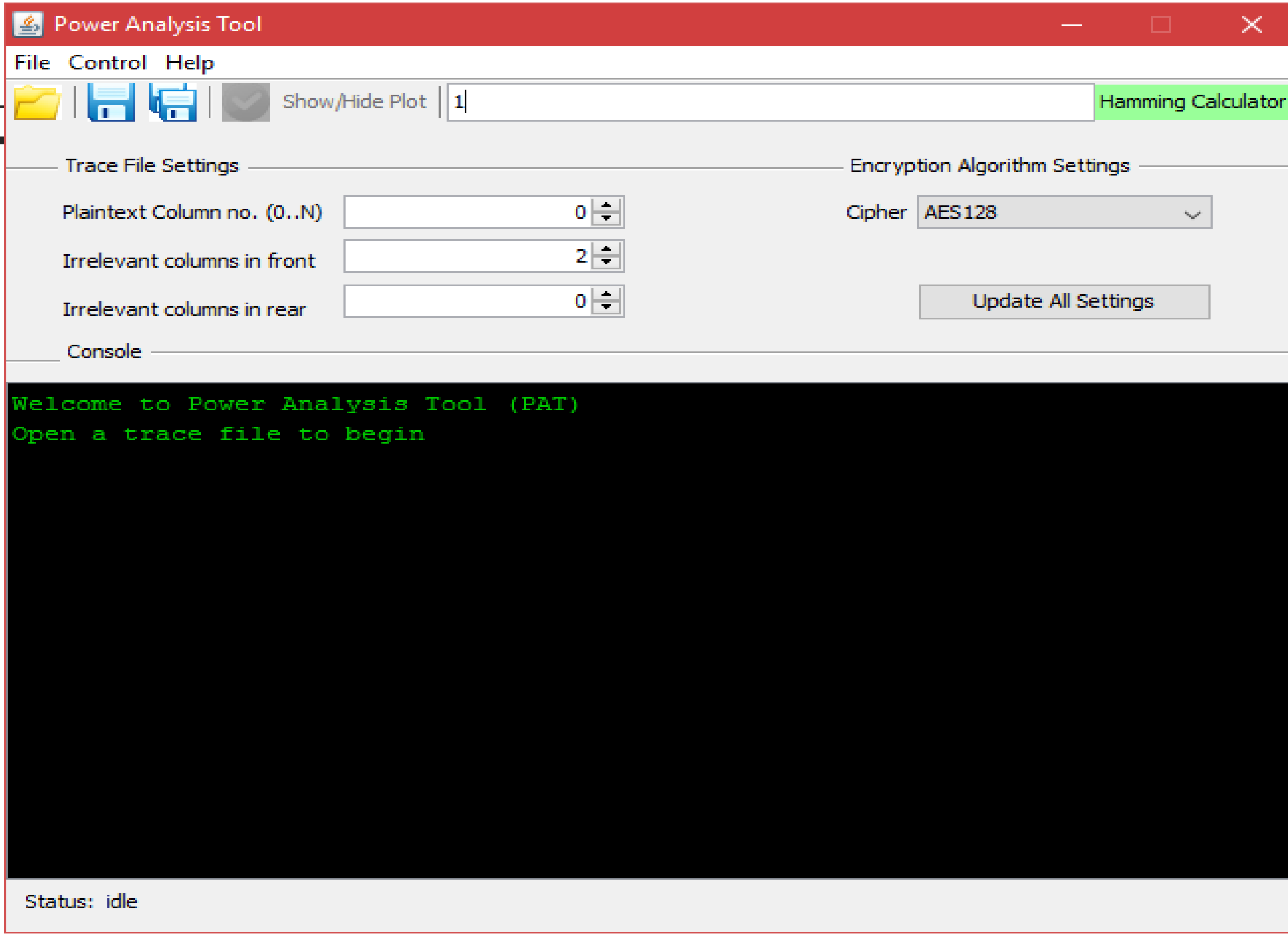
Power Analysis Tool (PAT)



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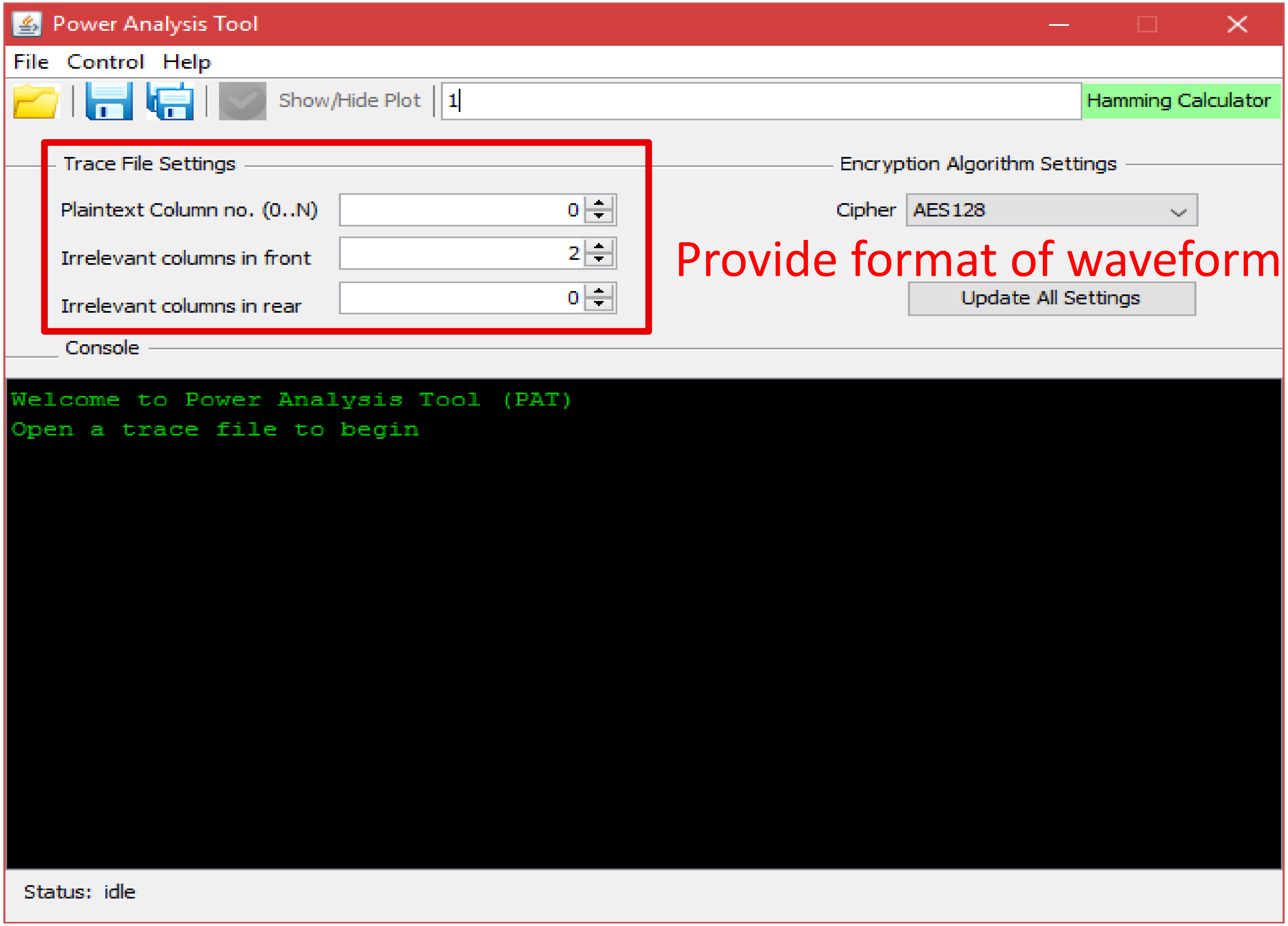
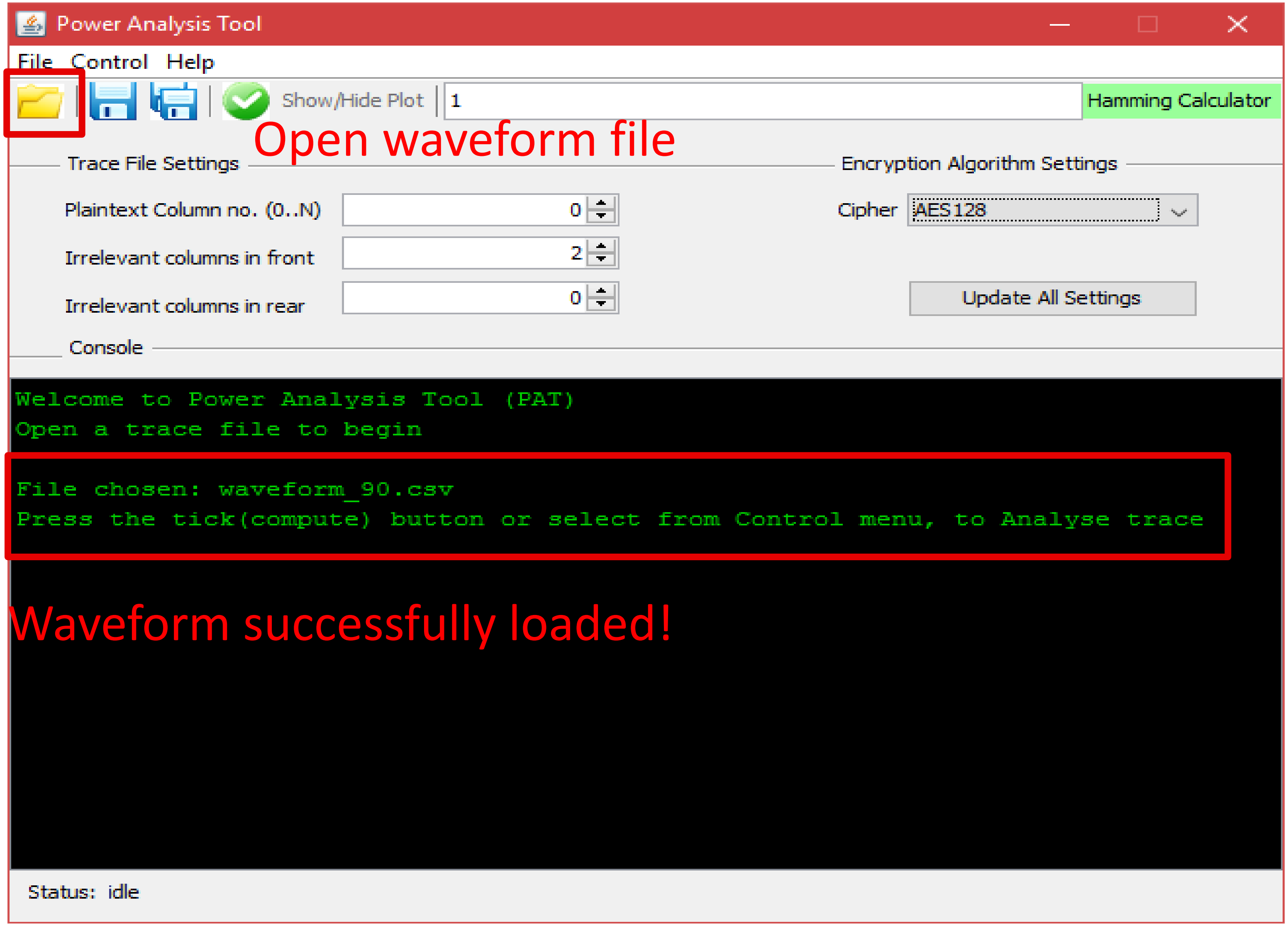


Provide format of waveform data

Power Analysis Tool (PAT)



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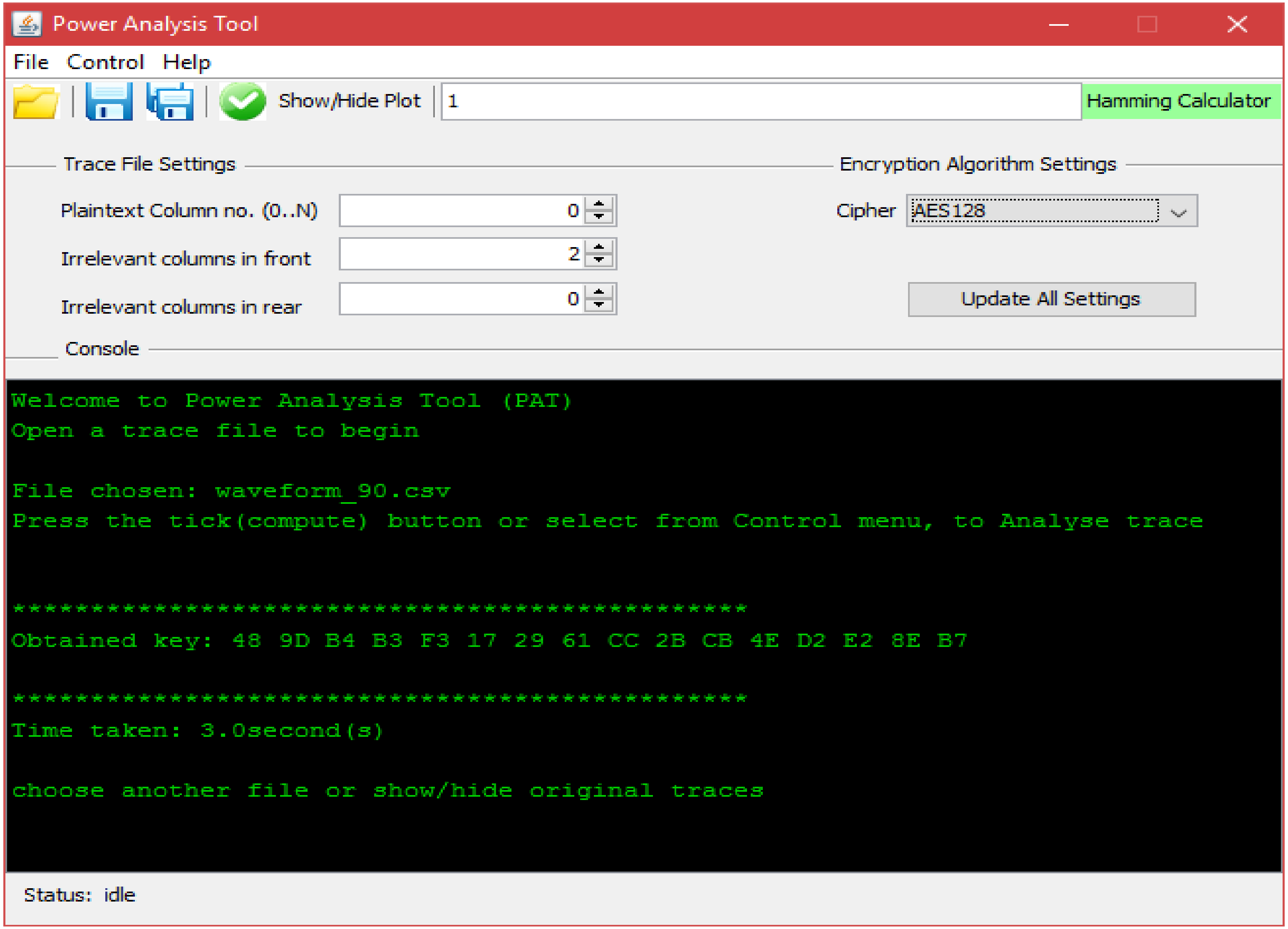

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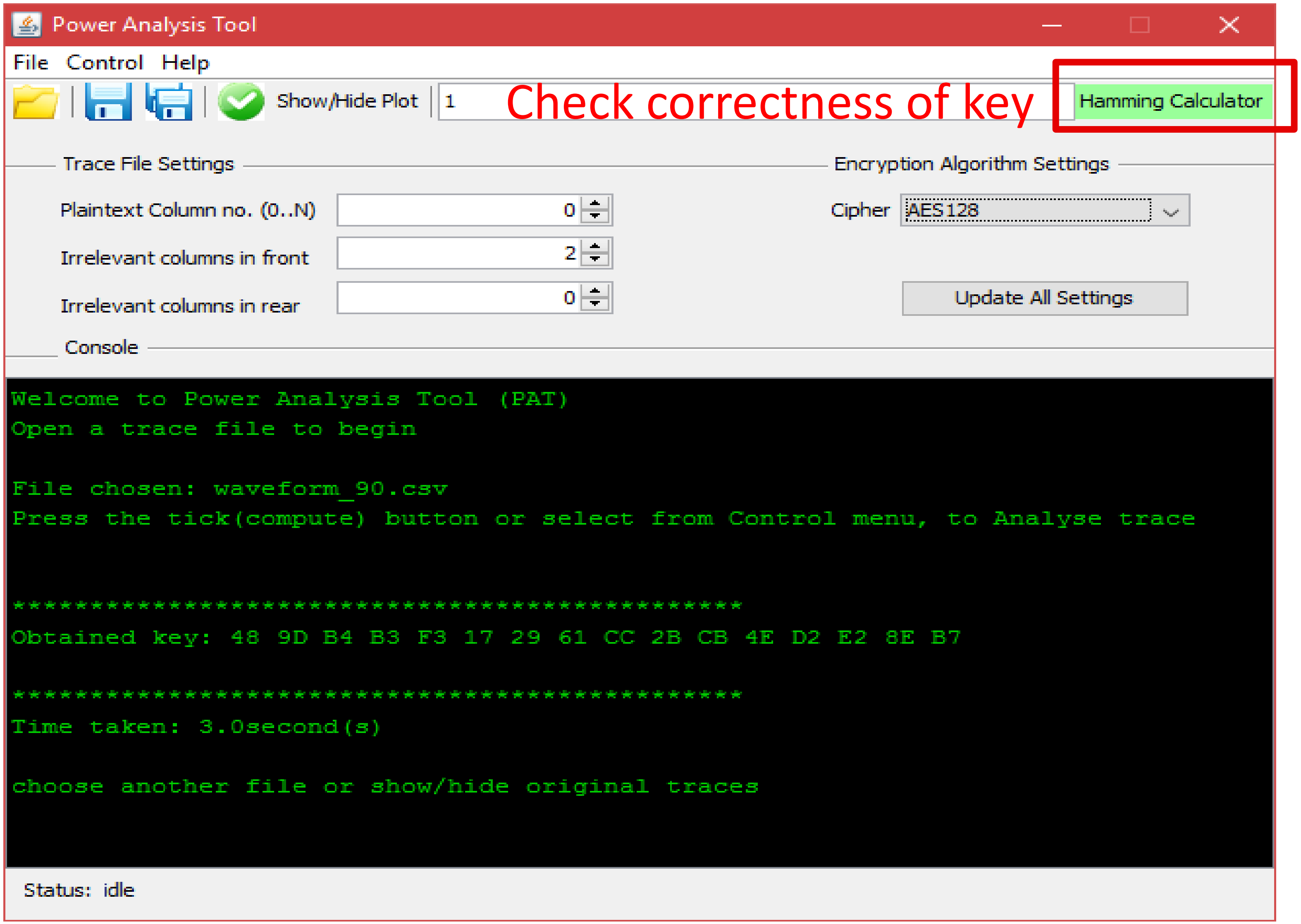
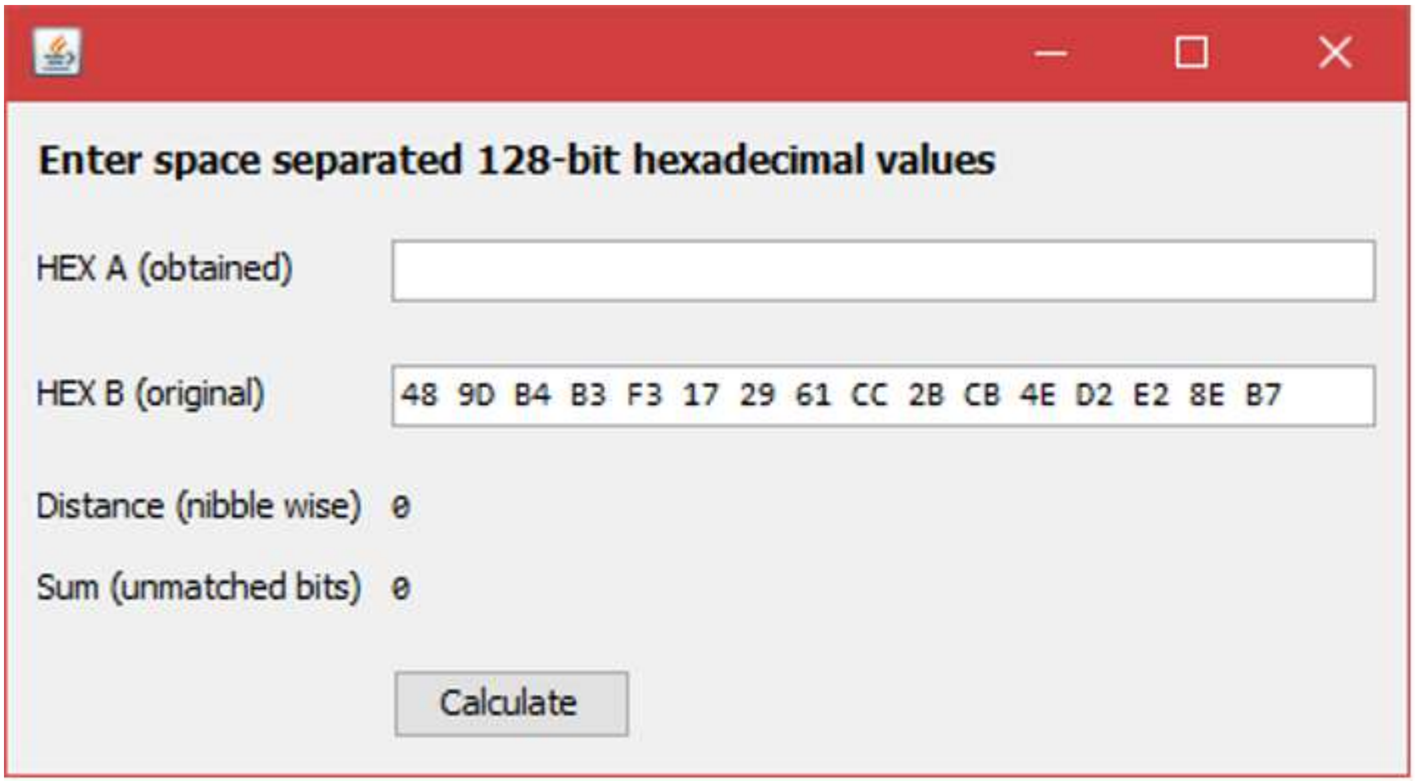


Power Analysis Tool (PAT)

Hamming Weight Calculator



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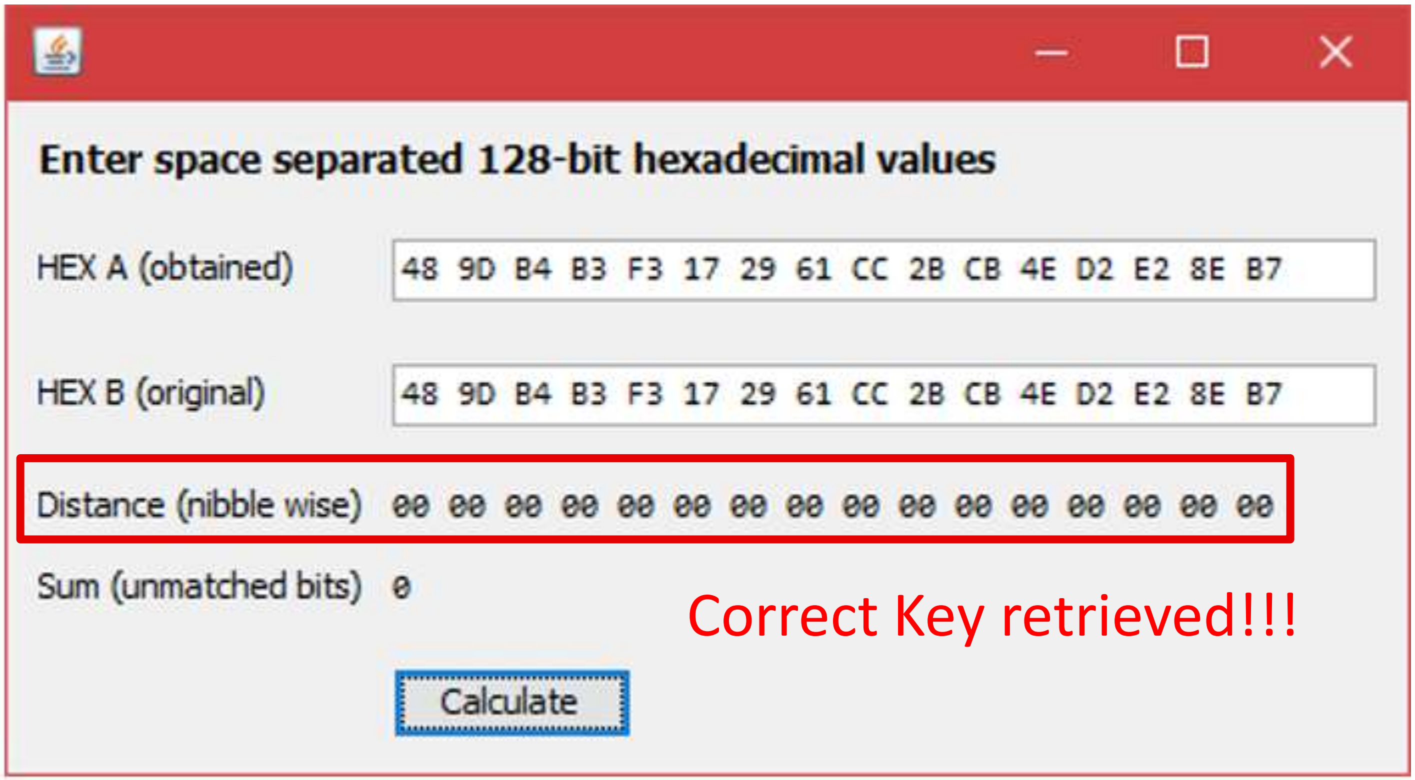

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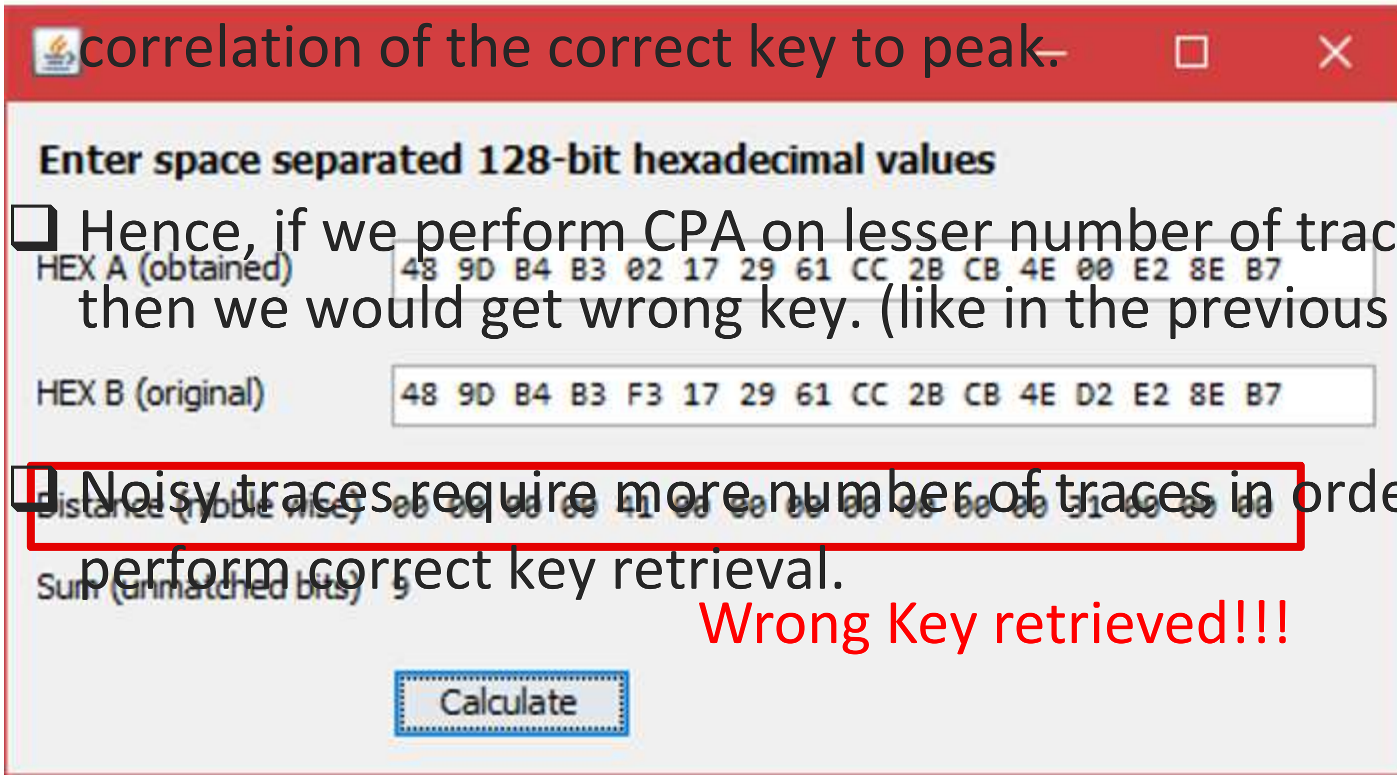
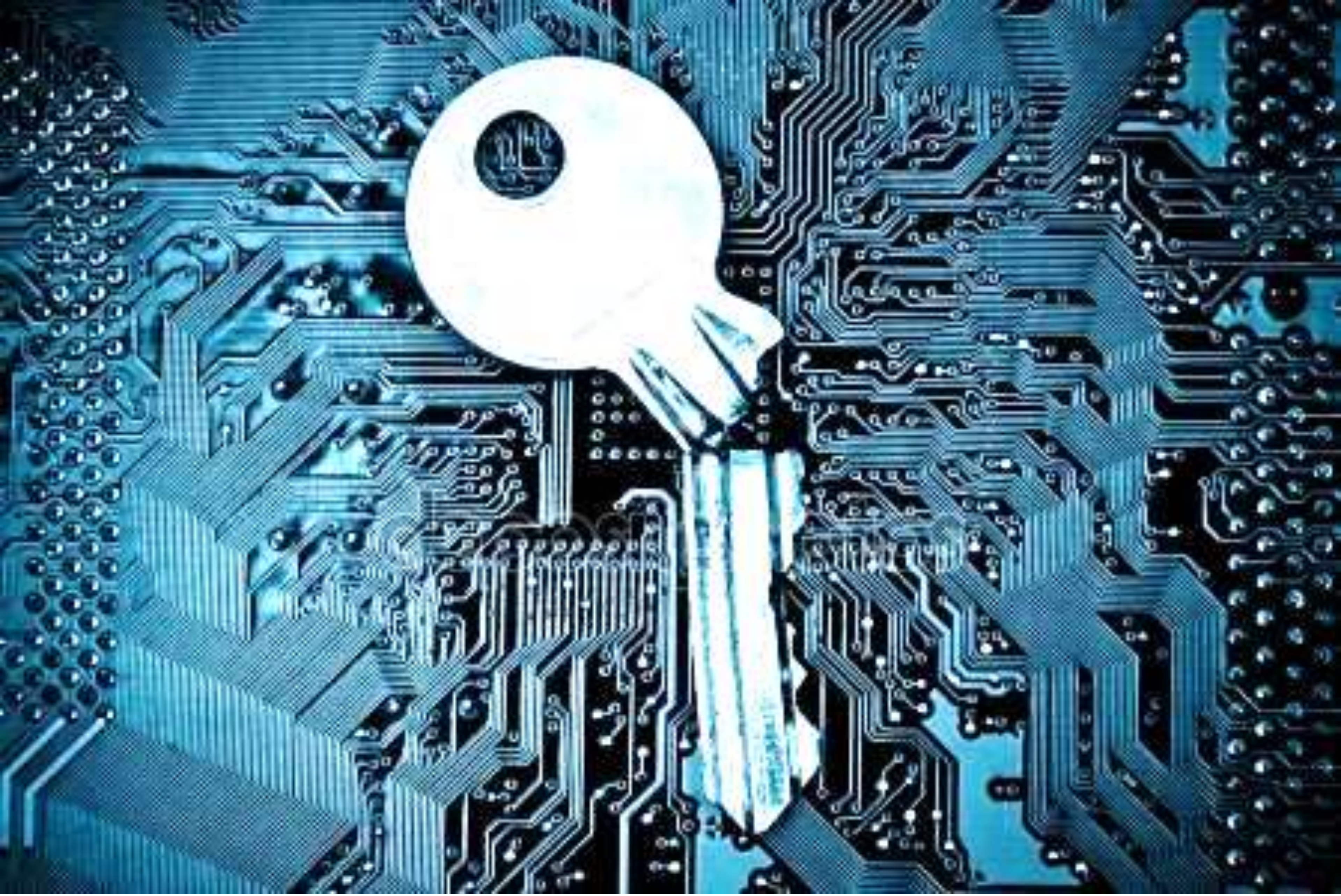


- It takes a certain minimum number of traces for the correlation of the correct key to peak.
- Hence, if we perform CPA on lesser number of traces, then we would get wrong key. (like in the previous case)
- Noisy traces require more number of traces in order to perform correct key retrieval.

Wrong Key retrieved!!!



Thank you